

# Simple Statistics and Archaeological Problems

James R. Allison  
Brigham Young University

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Among Keith Kintigh's many contributions to archaeology was his emphasis on understanding the connections among quantitative methods, archaeological problems, and what archaeologists can reasonably infer from their data. In both publications and in the classroom, he demonstrated the value of simple computer simulations to understand quantitative measures and how they behave when applied to actual archaeological data. Archaeological research increasingly incorporates analysis of large databases and quantitative methods appropriate to "big data", but simple statistics are still important to archaeological research. Even though archaeologists are increasingly using relatively sophisticated quantitative methods, there still are simple methods that are commonly misunderstood and misused (or sometimes just not applied when they would be helpful).

In this paper I want to take a close look at some simple methods: chi-square tests and related methods for analyzing contingency tables of cross-classified data. I use computer-intensive methods of resampling both to look at the distributions of several statistics in order to better understand them, and to suggest some ways in which resampling can help with interpretation in specific cases. Computer simulations provide a better understanding of the issues involved, and of how to use these (not always so) simple statistics to provide valid insights about archaeological questions.

My examples draw on data on obsidian sourcing from two Fremont sites in Utah Valley, Utah (**slide**): Woodard Mound and Wolf Village. These sites are only about 5 km apart but show some differences in the sources represented.

To set up the problem archaeologically:

The sites are both in southern part of Utah Valley, and date to between about A.D. 1000 and the 1200s

Obsidian at these sites comes from six different sources (slide (only five of the sources are on the map): the majority comes from either the Black Rock source or Topaz Mountain, sources that are approximately equidistant from the sites (about 105 km to Topaz Mountain, 110-115 to Black Rock).

More distant sources include Wildhorse Canyon, located south of Black Rock, and three sources in Idaho (Malad, Brown's Bench, and Bear Gulch) that I have lumped in my tables as Northern Sources (Malad is the most common)

The table includes samples of 100 obsidian tools and 100 pieces of debitage from each of the sites, 400 specimens total (I actually have more data than this but took random samples of 100 from each of the (column) categories).

My goals are largely methodological – but there are a couple of substantive questions that may be relevant. Assemblages of obsidian tools often contain more pieces from distant sources than debitage assemblages from the same site. There is some evidence in the table that this is true for both of these sites, but the counts are small for the distant sources and it is not immediately clear if the differences in the table are meaningful, or just sampling error. Also, there are differences in the sites: Wolf Village is larger, is positioned on a natural route into Utah Valley from the south, and was clearly a community center. It has communal architecture (slide), including this 80 m<sup>2</sup> pit structure, and there is good reason to expect that it might have more obsidian from distant sources .

(back to the table)

This kind of table is common in archaeology, with column (or row) variables related to provenience, and artifacts classified in various ways.

In the abstract, there are (at least) three kinds of things that you might want to know about that table of numbers:

1. is there a significant association between the row and column variables? (provenience and classification)
2. how strong is the association?
3. what is the structure/nature of the association?

Each of these is useful to look at, and resampling provides valuable insights into each of them. I'm going to run through a lot of things too fast, but I need to remember to say that the resampling here is done in R, (the graphs too). I use a couple of different methods of resampling, but mostly I'm resampling from a multinomial distribution based on the expected values (as you would calculate them in a chi-square test  $[r * c/g]$ ), but I also do some bootstrapping, where I resample from a multinomial distribution based on the observed values (rather than the expected values). In each case, I created 10,000 replicates.

## **Significance**

Significance is usually measured with a chi-square test (whether or not you do a formal chi-square test, it is often useful to have some sense of how likely any apparent association is to be just the result of sampling error)

I won't go far into the details of the test, but it involves calculating a statistic we call chi-square. That statistic is then compared to a chi-square distribution (the fact that the statistic and the distribution have the same name can be confusing). Basically if there is no association between rows and columns (provenience and obsidian source, in this case) then, if you take a bunch of

random samples of the same size and calculate the chi-square statistic for each of those samples, the sampling distribution of the statistic will be approximately the same as a chi-square distribution with the appropriate number of degrees of freedom (rows-1 times columns - 1), in this case 9 degrees of freedom.

In archaeology, including in my example, we are often interested in differences in assemblage composition between proveniences. We can (often) observe these differences in our samples, and we would like to know if they reflect real (and important) differences in the populations the samples are drawn from. Because we can calculate the chi-square distribution, it provides a short-cut--- you don't actually need to take a bunch of random samples to get an estimate of the chances that patterns in your data are simply due to sampling error.

In the case of my table, Chi-square = 22.4, if you compare the chi-square statistic to the theoretical chi-square distribution to get an estimate of the p-value,  $p = .008$ , or about 8 chances in a thousand of getting differences as large as the ones in the data in random samples from. Remember it is only an estimate, though often a good one, because the chi-square distribution is only an *approximation* to the sampling distribution of the chi-square statistic.

If you actually resample the table (slide) you can check how good the approximation is – in this case pretty good (.008 of the resampled chi-square values are larger than the one we had for the original sample). But, in other cases, depending on the details of the table, the approximation may not be very good (low expected values and low total counts may be problems). Resampling provides a good check on the standard test.

Even here, in a relatively well-behaved case, there are a couple of things to pay attention to (slide). Notice, for instance, how bumpy the resampling results are in the right tail of the distribution. This means that it probably isn't a good idea to make too much of the precise value

of the estimated probability in a standard chi-square test, and especially not when the probabilities are small.

### **Strength of Association**

Resampling becomes even more useful for looking at the strength of the association.

Significance in the chi-square test depends on a combination of the sample size and how strong the association between the row variables and column variables is. It may be easier to think about association in terms of differences between the sites and artifact types in the abundance of sources. There are a number of statistics that measure strength of association; the most commonly used one is Cramer's V, which is a transformation of the chi-square statistic: you divide out the sample size and correct for the number of cells in the table. This scales Cramer's V to between zero and one, with larger values indicating stronger association. But, while Cramer's V gets reported often enough, it's rare to see anyone offer a useful interpretation. I think that is largely due to the fact that Cramer's V is not in any sensible units – larger values mean a stronger association, but that's usually about as much as you can say.

In the original table, for instance, Cramer's V is .137. So what does that mean? Does the low value mean that the association between variables is weak? Is it a strong enough association to care about? It just isn't clear.

More interestingly, if we split the original table by site, we get two smaller tables that compare the abundance of sources in debitage with their abundance in the formal tools.

Wolf Village Cramer's V = .204, Woodard Mound = .231

But what does that mean? More precisely, can we interpret the larger Cramer's V value at Woodard Mound as indicating an important difference between the two sites?

Resampling, in this case bootstrapping based on the observed values (not the expected values like before) provides some insight (slide).

95% confidence for Wolf Village .109-.338

95% confidence for Woodard Mound .133-.349

I'm still not sure I know whether that is a strong association, but at least we can say the difference in strength of association between the two sites does not seem to be important, as it easily could result from sampling error alone.

### **The Structure of the Table**

But none of this has gotten to the really interesting questions here, of what actually is going on (slide). What differences are driving the significant chi-square and the strength of association in this dataset? There are a number of ways to get at this, starting with simple methods like just looking at percentages (in this table the counts are also the column percentages, which makes things easy). We can notice some differences, but it is difficult to know which differences are important and which can just be due to sampling error.

One way of getting at this (recommended in several statistics books for archaeology) is to use something called adjusted residuals (slide). I won't go into the calculations, but they calculate how much individual cell values deviate from expected values, and then are adjusted so that they (theoretically) have a sampling distribution that is approximately normal, with a mean of zero and a standard deviation of one. That means they can be interpreted as z-scores, and you can easily find the probability of the value of any cell in the table deviating from the expected value as much as (or more than) the observed value in the sample. It is common to look at a table of adjusted residuals and highlight those larger than 1.96 or smaller than -1.96, because those z-

scores are associated with probability of .05, (which unfortunately has become a magic number). Here I've actually highlighted a few more of the largest residuals; these are the cells that deviate from expected values in ways that are worth paying attention to.

My resampling (slide) suggests that the distributions of the adjusted residuals are actually sometimes a bit squirrely and not always normally distributed, but they always have a mean of zero and a standard deviation of one. The deviation from normal (slide) is especially notable for the cells with the lowest expected values (which is what you would expect), but even there the mean and standard deviation are what they should be.

I'm not convinced this is the best way to think about the problem, however. In the resampled tables, the same cell value can have a different adjusted residual depending on what the other values in the table are (because the expected value changes). It seems more intuitive to me (and simpler) to look at things one cell at a time and ask (for example) how likely a value of 31 is in the upper left cell of the table. Fortunately, there are a couple of ways of getting at that.

One way is to think of the question as fundamentally a binomial problem. Taking one cell at a time, every piece of obsidian in my table is (for example) either debitage from Woodard Mound or it isn't. It is then possible to use the normal approximation to the binomial distribution to calculate z-scores analogous to the adjusted residuals (slide). Or (slide) you can take the more direct (but computationally more difficult) approach, and simply calculate the probability of your observed value, given the expected probability of success for that cell (which is simply the expected counts divided by the grand total of the table). Finally, you could just resample your original table a bunch of times and estimate the probability of your observed values from the results (slide). All of these methods provide similar results, although the exact numbers differ slightly (in particular, the smallest probabilities are even smaller in the resampling).



## Conclusions

The substantive conclusions aren't really the point here but, based on this analysis (slide), we can say that there is significantly less distant obsidian in the debitage than in the tools. There also is significantly more Topaz Mountain obsidian than expected in the Woodard Mound debitage, and less in the Wolf Village tools, as well as unusual amounts of obsidian from the distant northern sources to the north in the Wolf Village tools.

There are two methodological points that I hope I have made (both of which I originally learned from Keith Kintigh). First, resampling is a powerful tool for understanding what is going on with the statistical methods we use, and it is particularly useful for things like finding confidence intervals for the Cramer's V statistic where (to my knowledge) there is no analytical solution. Finally, I'll just agree with this quote (slide): "what we hope is for the archaeologist to realize that there is usually no single best method, and that a variety of approaches must be used to gradually gain an understanding of the structure of the data and its relevance to the anthropological questions" (Kintigh 1987).



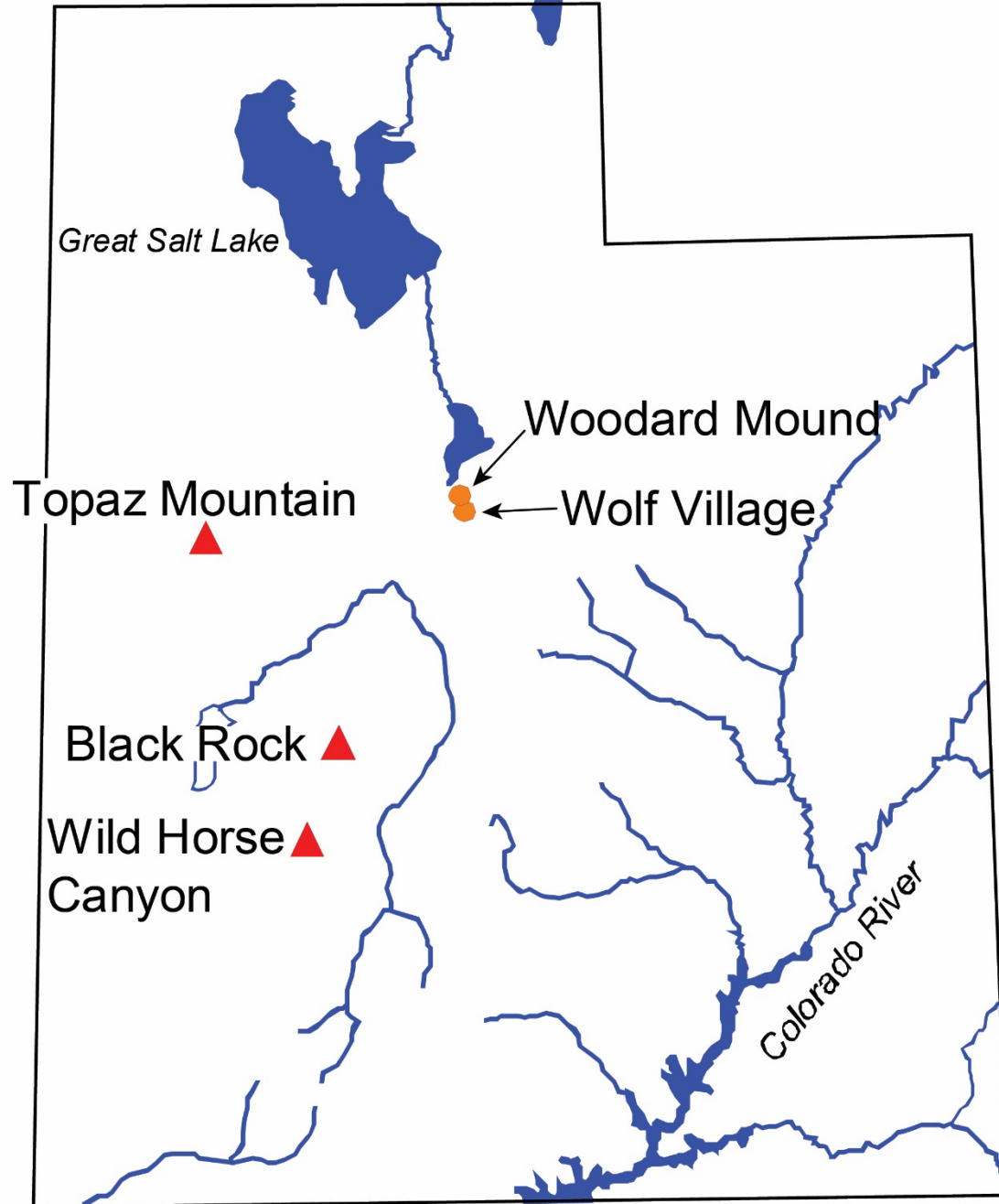
Woodward Mound

Utah Lake

Wolf Village

▲ Brown's Bench

▲ Malad





# Obsidian from Wolf Village and Woodard Mound

|                  | Woodard Mound |       | Wolf Village |       |       |
|------------------|---------------|-------|--------------|-------|-------|
| Source           | Debitage      | Tools | Debitage     | Tools | Total |
| Black Rock       | 31            | 32    | 35           | 39    | 137   |
| Topaz            | 66            | 52    | 56           | 44    | 218   |
| Wildhorse Canyon | 1             | 7     | 6            | 4     | 18    |
| Northern Sources | 2             | 9     | 3            | 13    | 27    |
| Total            | 100           | 100   | 100          | 100   | 400   |
|                  |               |       |              |       |       |



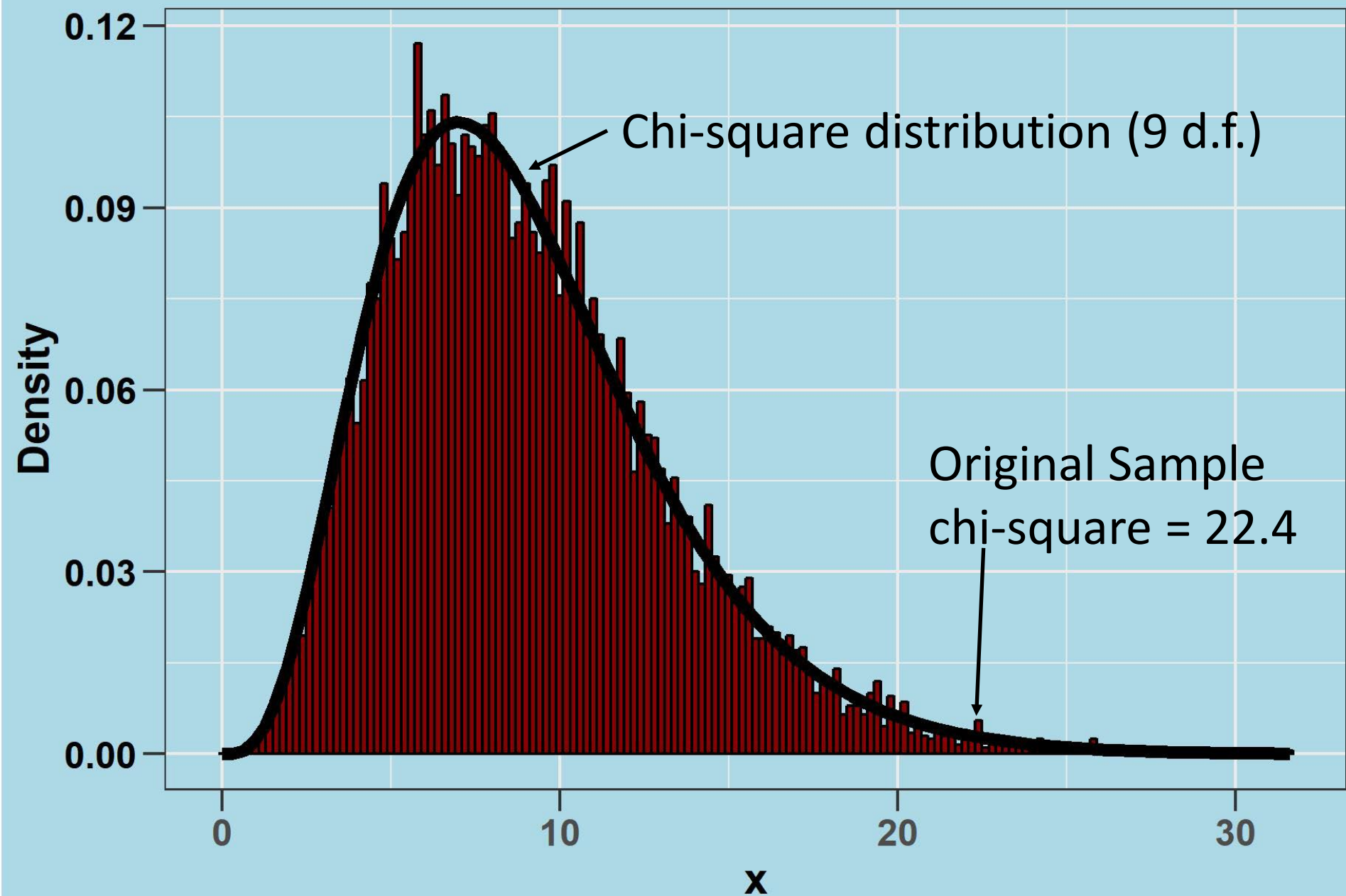




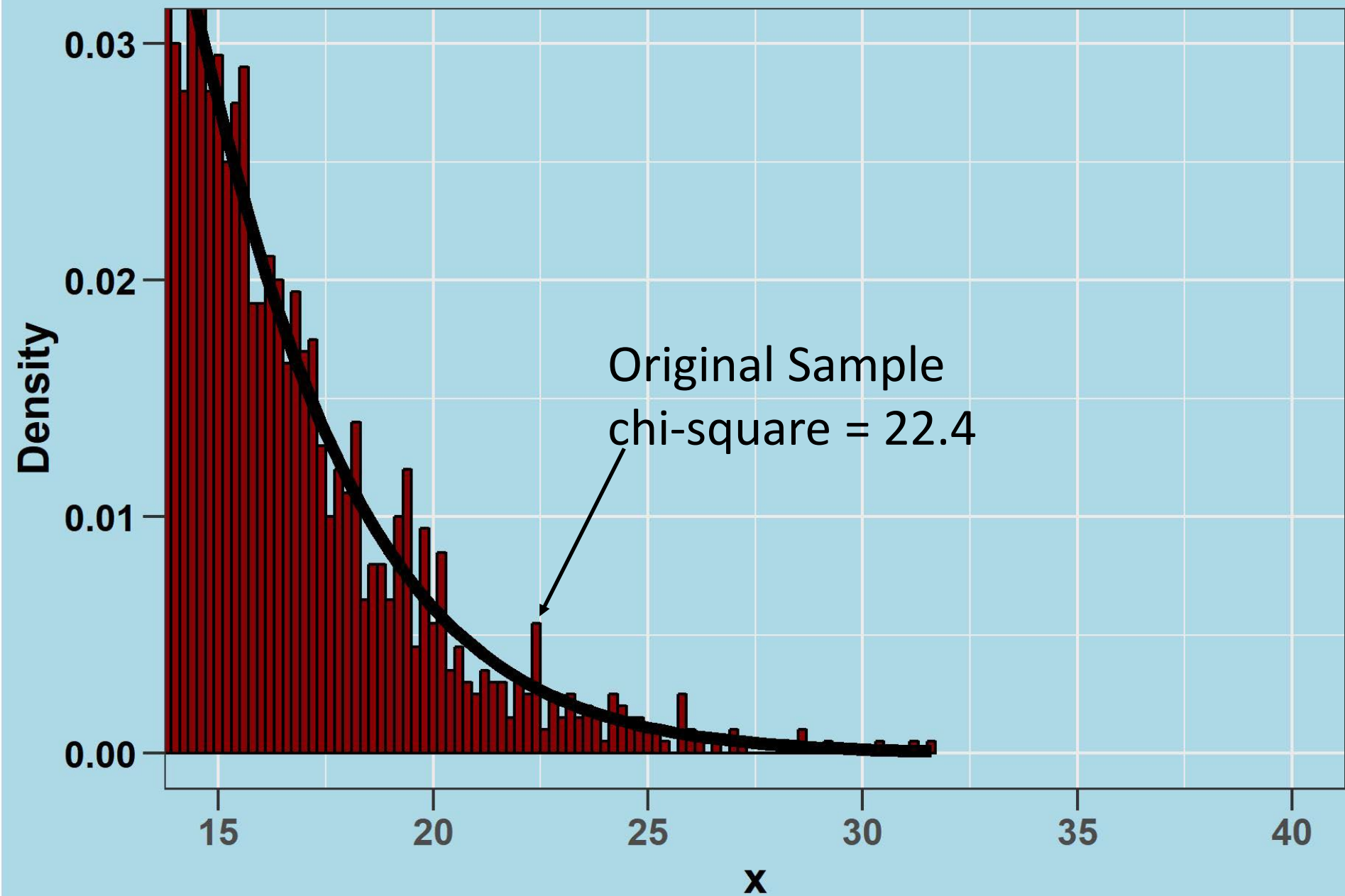
# Obsidian from Wolf Village and Woodard Mound

|                  | Woodard Mound |       | Wolf Village |       |       |
|------------------|---------------|-------|--------------|-------|-------|
| Source           | Debitage      | Tools | Debitage     | Tools | Total |
| Black Rock       | 31            | 32    | 35           | 39    | 137   |
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|                  |               |       |              |       |       |

# *Chi-Square Values*



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|                  | Woodard Mound |       |
|------------------|---------------|-------|
| Source           | Debitage      | Tools |
| Black Rock       | 31            | 32    |
| Topaz            | 66            | 52    |
| Wildhorse Canyon | 1             | 7     |
| Northern Sources | 2             | 9     |
| Total            | 100           | 100   |

Woodard Mound  
Cramer's V = .231

|                  | Wolf Village |       |
|------------------|--------------|-------|
| Source           | Debitage     | Tools |
| Black Rock       | 35           | 39    |
| Topaz            | 56           | 44    |
| Wildhorse Canyon | 6            | 4     |
| Northern Sources | 3            | 13    |
| Total            | 100          | 100   |

Wolf Village  
Cramer's V = .204

# *Bootstrapped Cramer's V*



# Obsidian from Wolf Village and Woodard Mound

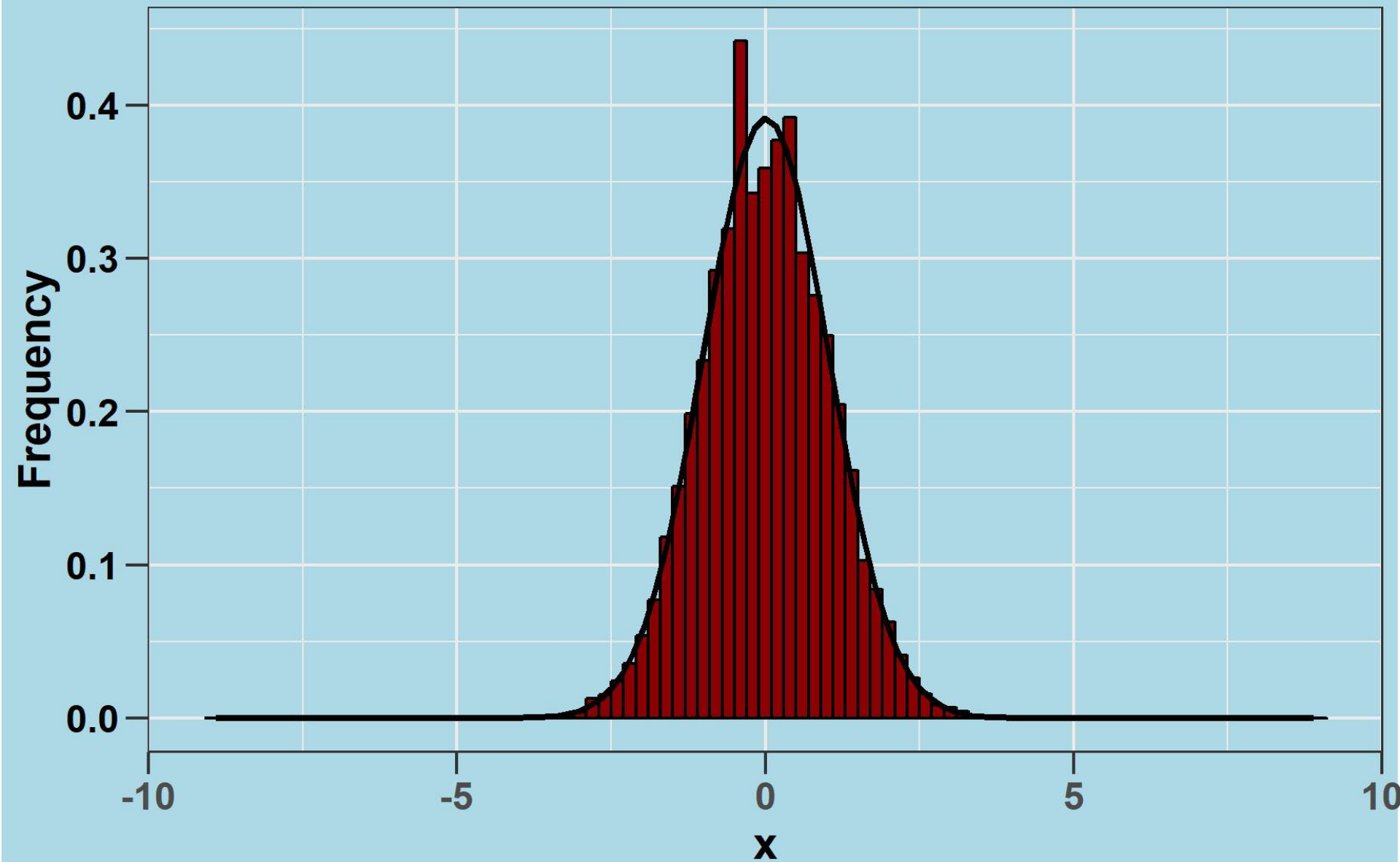
|                  | Woodard Mound |       | Wolf Village |       |       |
|------------------|---------------|-------|--------------|-------|-------|
| Source           | Debitage      | Tools | Debitage     | Tools | Total |
| Black Rock       | 31            | 32    | 35           | 39    | 137   |
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| Northern Sources | 2             | 9     | 3            | 13    | 27    |
| Total            | 100           | 100   | 100          | 100   | 400   |
|                  |               |       |              |       |       |

# Adjusted Residuals

|                  | Woodard Mound |       | Wolf Village |              |
|------------------|---------------|-------|--------------|--------------|
| Source           | Debitage      | Tools | Debitage     | Tools        |
| Black Rock       | -0.79         | -0.55 | 0.18         | 1.16         |
| Topaz            | <b>2.67</b>   | -0.58 | 0.35         | <b>-2.43</b> |
| Wildhorse Canyon | <b>-1.95</b>  | 1.39  | 0.84         | -0.28        |
| Northern Sources | <b>-2.19</b>  | 1.04  | <b>-1.73</b> | <b>2.88</b>  |

# ***Adjusted Residual[1,1]***

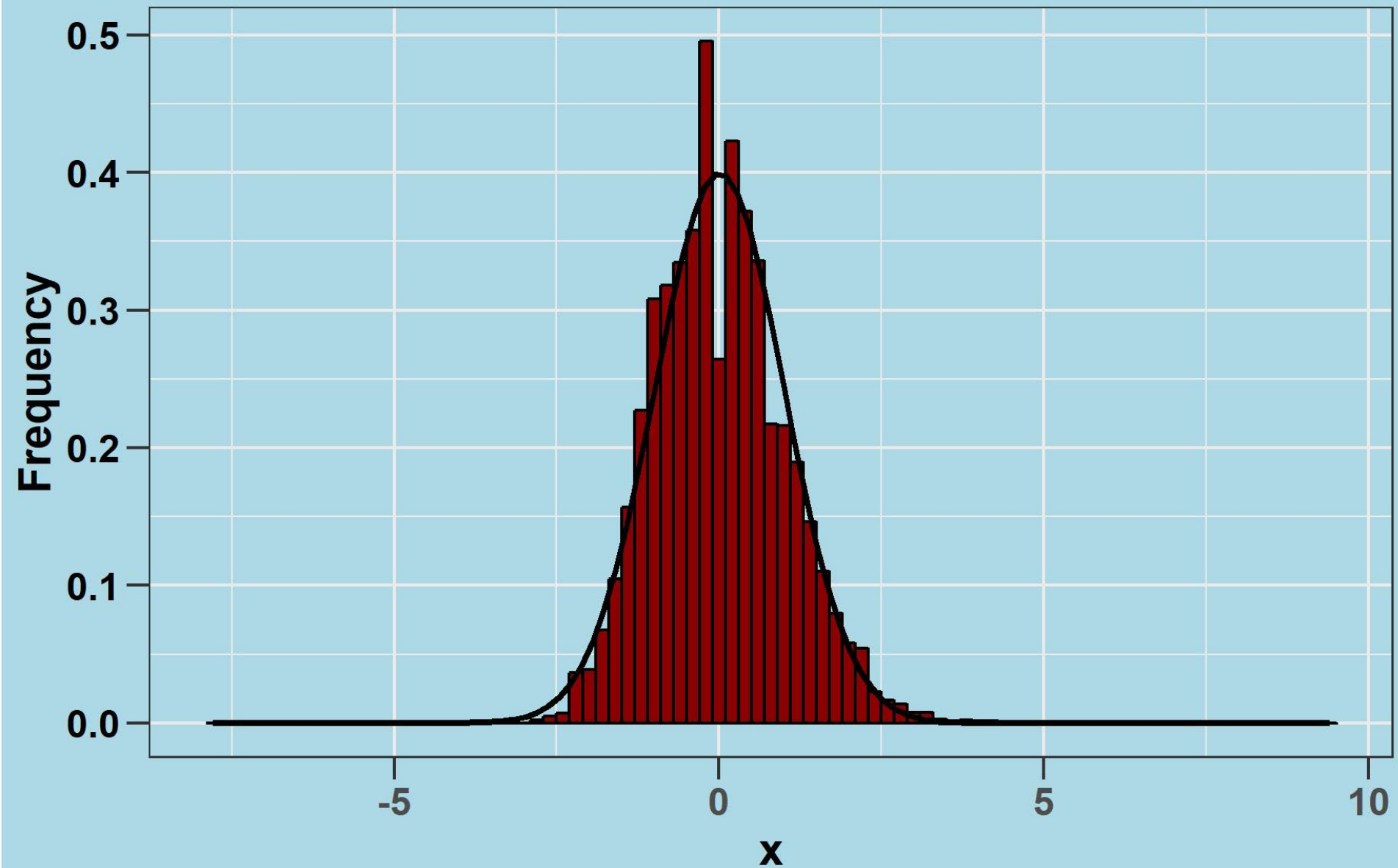
***mean = 0 , s.d. = 1***





# ***Adjusted Residual[3,1]***

***mean = 0 , s.d. = 1***



# Z-Scores Based on Normal Approximation to the Binomial Distribution

|                  | Woodard Mound |       | Wolf Village |              |
|------------------|---------------|-------|--------------|--------------|
| Source           | Debitage      | Tools | Debitage     | Tools        |
| Black Rock       | -0.58         | -0.40 | 0.13         | 0.85         |
| Topaz            | <b>1.68</b>   | -0.36 | 0.22         | <b>-1.53</b> |
| Wildhorse Canyon | <b>-1.66</b>  | 1.19  | 0.71         | -0.24        |
| Northern Sources | <b>-1.84</b>  | 0.87  | <b>-1.46</b> | <b>2.43</b>  |

Probabilities of Observed Cell Values  
Based on the Binomial Distribution  
(red means observed is < expected)

|                  | Woodard Mound |       | Wolf Village |       |
|------------------|---------------|-------|--------------|-------|
| Source           | Debitage      | Tools | Debitage     | Tools |
| Black Rock       | 0.32          | 0.39  | 0.47         | 0.22  |
| Topaz            | 0.06          | 0.39  | 0.44         | 0.07  |
| Wildhorse Canyon | 0.06          | 0.17  | 0.30         | 0.53  |
| Northern Sources | 0.03          | 0.24  | 0.09         | 0.02  |

Probabilities of Observed Cell Values  
Based on Resampling  
(red means observed is < expected)

|                  | Woodard Mound |       | Wolf Village |       |
|------------------|---------------|-------|--------------|-------|
| Source           | Debitage      | Tools | Debitage     | Tools |
| Black Rock       | 0.28          | 0.36  | 0.48         | 0.18  |
| Topaz            | 0.01          | 0.34  | 0.43         | 0.02  |
| Wildhorse Canyon | 0.06          | 0.17  | 0.29         | 0.53  |
| Northern Sources | 0.03          | 0.23  | 0.09         | 0.02  |

## Back to the Original Table

|                  | Woodard Mound |       | Wolf Village |       |       |
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“...what we hope is for the archaeologist to realize that there is usually no single best method, and that a variety of approaches must be used to gradually gain an understanding of the structure of the data and its relevance to the anthropological questions.”

Kintigh 1987

“Quantitative Methods Designed for Archaeological Problems”